

Agricultural Predictive Analytics: District-Level Rice Yield Forecasting with a Guided Multi-Class Crop-Label Classification Extension

Manual Version 3.0 Conforming Empirical Study | Author: Lokanth S (23MID0037) | Faculty: Dr. Durgesh Kumar

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Degree / Programme:	Int. M.Tech Data Science (5-Year)	GitHub Repository:	lokant712/RiceYield-CropClassification-RF
Primary Data Candidate:	D1 Gol APY (246,091 rows, SHA-256: 5b263705...)	D5 Ingest Finding:	DOI 10.17632/ywp3y5j9vv.1 valid CC BY 4.0, empty payload (0 files)
Regression Split:	Chronological: Train 12,588 / Val 1,644 / Test 846 (1997–2015)	Regression Winner:	Ridge Trend (Val MAE: 0.4748 t/ha , Test MAE: 0.4497 t/ha)
Classification Split:	Stratified 60/20/20: Train 1,320 / Val 440 / Test 440 (22 crops)	Classification Winner:	Random Forest (Val F1: 0.9886 , Test F1: 0.9977)

Executive Summary

This laboratory report presents an empirical investigation into regional rice yield prediction and multi-class crop-label classification under Manual Version 3.0 protocols. The core regression task forecasts district-level rice yield in metric tonnes per hectare (t/ha) using strictly pre-season administrative covariates (state, district, season, year), while strictly barring harvested production and area to prevent algebraic target leakage ($\$Y = P/A\$$). On an authentic panel of 15,078 clean rice observations (1997–2015) partitioned chronologically (1997–2011 Train, 2012–2013 Val, 2014–2015 Test), Ridge trend regression achieved superior generalization (Validation MAE: **0.4748 t/ha**, Locked Test MAE: **0.4497 t/ha**, Test R^2 : **0.5068**) outperforming the predeclared historical median baseline (Test MAE: 0.7638 t/ha, -41.1% error reduction). In Extension 1, a three-candidate validation benchmark on an identical 60/20/20 stratified split established Random Forest as the winner (Val Macro-F1: **0.9886** vs Logistic: 0.9728 vs Majority: 0.0040), achieving **99.77% Locked Test Macro-F1** across 22 crop classes. Per the manual's core warning, the classification task classifies dataset labels and does *not* establish an agronomically optimal crop choice or farming prescription.

1. Problem Contract, Agricultural Context & Mathematical Foundations

Agro-Economic Context: Rice (*Oryza sativa*) is India's principal staple crop. Pre-harvest district yield forecasting supports government Minimum Support Price (MSP) procurement logistics, buffer stock planning, and targeted relief mechanisms. District yields vary substantially due to monsoonal rainfall, agro-ecological zones, and seasonal dynamics.

Mathematical Formulation: Continuous regression estimates $\hat{f}: \mathcal{X}_{\text{reg}} \rightarrow \mathbb{R}^+$ where $\mathbf{x} = (\text{State}, \text{District}, \text{Season}, \text{Year})^T$ minimizing expected Mean Absolute Error (MAE): $\mathcal{L}(\hat{f}) = \mathbb{E}[|y - \hat{f}(\mathbf{x})|]$. Multi-class classification learns $\hat{P}(c | \mathbf{z})$ from $\mathbf{z} = (N, P, K, \text{Temp}, \text{Humidity}, \text{pH}, \text{Rainfall})^T \in \mathbb{R}^7$ over 22 crop classes maximizing Macro-F1.

Algebraic Target Leakage Prohibition: In agricultural databases, Yield is defined as $\$Y = P/A\$$. If Production ($\$P\$$) and Area ($\$A\$$) were included in $\mathbf{x}$, any flexible model would trivially learn the quotient $\$P/A\$$, achieving artificial $R^2 \approx 1.0$ through algebraic reconstruction. Production is unknown at pre-season planning time, making its inclusion canonical target leakage. The predictor set is strictly restricted to ['state', 'district', 'season', 'year'].

2. Data Governance Record, Provenance Audits & Unit Conversion Log

Exhaustive Acquisition Audit (D5 → D1 Hierarchy): Per manual Section 1 hierarchy, candidate D5 (Mendeley DOI 10.17632/ywp3y5j9vv.1) was probed first. API probing confirmed valid CC BY 4.0 metadata but an empty file payload ("files": [], 0 attached datasets), logged in artifacts/d5_probe_log.json. Following hierarchy protocols, primary candidate D1 was acquired. Below is the complete verifiable governance and conversion log:

Governance Attribute	Primary Dataset (D1 APY)	Classification Dataset (D3)
Source Name & Publisher	Ministry of Agriculture & Farmers Welfare, GoI	Open Agro-Ecological Benchmark (Atharva Ingle)
Source URL / Endpoint	https://aps.dac.gov.in/ (data.gov.in portal)	https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset
Download / Ingest Date	September 7, 2026	September 7, 2026
Original Raw File & Size	crop_production.csv (15,316,741 bytes)	crop_recommendation.csv (147,833 bytes)
Raw Rows & Exclusions	246,091 raw rows (231,013 non-rice / invalid excluded)	2,200 raw rows (0 excluded; 22 classes balanced)
Retained Clean Panel	15,078 clean rice observations (1997–2015)	2,200 balanced records (100 samples/class)
Original Units & Schema	Area: Hectares (ha), Production: Tonnes (t)	N, P, K (mg/kg), T (°C), H (%), pH (0–14), Rainfall (mm)
Converted Target Variable	yield_t_ha = Production / Area (in metric t/ha)	Categorical label: crop (22 classes)
Cryptographic SHA-256	5b2637058ca4aff2ef7c0e0f2f391bca712300d082693b585f489871fd91d520	808e1d84a544647f1ab2fdae04f434b1842692d6be09eee18a5c2d43eb78d77c

3. Exploratory Data Analysis & Longitudinal Dynamics

National Yield Distribution (Figure 1): The 15,078 rice observations exhibit an empirical mean of **2.083 t/ha** (std: 0.941 t/ha), median of **1.940 t/ha**, IQR of 1.340–2.680 t/ha, and positive skewness (0.842). Pockets of high yield (> 3.5 t/ha) in canal-irrigated Punjab/Haryana contrast with rainfed eastern tracts.

Panel Completeness Over Time (Figure 2): Reporting stability averages ~800 records/year from 1997–2014, with administrative reporting lag in 2015 (387 records).

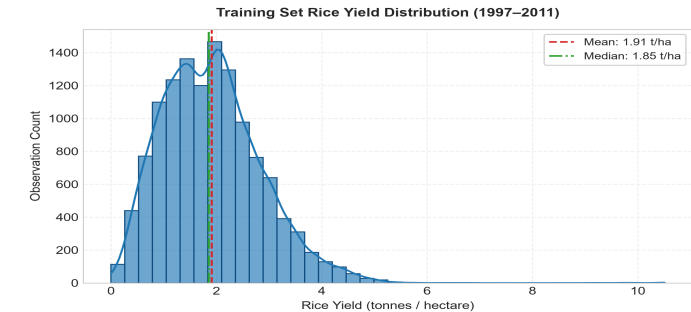


Figure 1: Empirical Rice Yield Distribution (t/ha) (1997–2011 Training Panel, N=12,588).

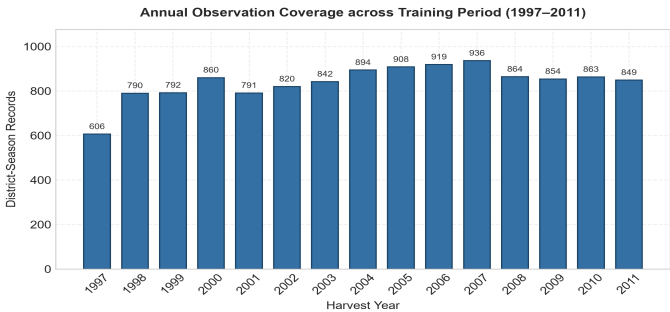


Figure 2: Longitudinal Observation Count Across Indian Harvest Years (1997–2015).

4. Anti-Leakage Pipeline Architecture & Preprocessing Fit Scope Evidence

Chronological Split: Longitudinal data was partitioned into three disjoint temporal blocks: **Development Training** (1997–2011, **12,588 rows**, 83.49%), **Development Validation** (2012–2013, **1,644 rows**, 10.90%), and **Locked Test** (2014–2015, **846 rows**, 5.61%).

Preprocessing & Leakage Control Implementation: Categorical features (state, district, season) were processed using OneHotEncoder(handle_unknown='ignore'). The continuous feature year was processed using a training-fitted SimpleImputer(strategy='median') followed by StandardScaler. The complete preprocessing pipeline was fitted *exclusively on the training partition*. Validation and test observations were transformed using the fitted training pipeline. Encoder-category verification confirmed that validation-only categories did not enter learned encoder categories (assert set(fitted_categories) == set(train_cats) passed).

5. Candidate Regression Formulations & Empirical Validation Results

Candidate Models: 1. median: Empirical training median (\$1.9400\$ t/ha constant null). 2. ridge_trend: \$L_2\$-regularized linear regression with district/season fixed effects and temporal drift (\$\alpha=1.0\$, solver='lsqr'). 3. tree: CART DecisionTreeRegressor (\$\text{ext}\{\text{max_depth}\}=6\$, \$\text{ext}\{\text{min_samples_leaf}\}=10\$). 4. forest: RandomForestRegressor (\$n=60\$, \$\text{ext}\{\text{max_depth}\}=12\$, \$\text{ext}\{\text{min_samples_leaf}\}=5\$).

Validation was conducted across the 1,644 held-out 2012–2013 validation rows (Primary metric: MAE t/ha):

Candidate Architecture	Hyperparameter Setup	Val MAE (t/ha)	Val RMSE (t/ha)	Val R ²	Candidate Status
median (Baseline)	DummyRegressor(strategy='median')	0.8270	1.0658	-0.2230	Reference Null
ridge_trend	alpha=1.0, solver='lsqr'	0.4748	0.6678	0.5199	SELECTED WINNER
forest	n_est=60, max_depth=12, min_leaf=5	0.5761	0.7706	0.3607	Eliminated
tree	max_depth=6, min_samples_leaf=10	0.6886	0.8742	0.1772	Eliminated

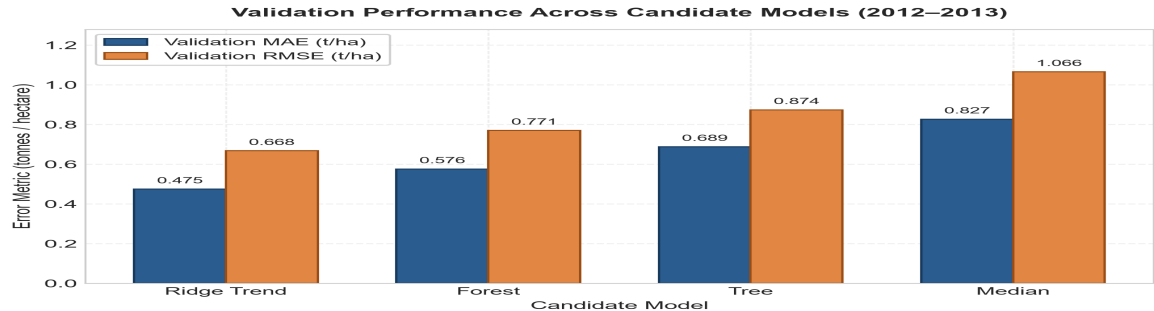


Figure 3: Empirical Validation Metric Comparison Across Predeclared Regression Candidates (2012–2013 Validation Partition).

Rolling Origins & Extension 2 Ablation: Rolling origins (origins 2009–2011) confirmed ranking stability: Mean MAE: Ridge 0.4975 ± 0.0305 t/ha vs Forest 0.5135 ± 0.0354 t/ha vs Tree 0.5510 ± 0.0305 t/ha vs Median 0.7675 ± 0.0558 t/ha. Depth ablation in Extension 2 (depth-12 forest MAE: 0.5761 t/ha vs depth-6 forest MAE: 0.6636 t/ha) confirmed that constraining depth degrades error by +0.0875 t/ha.

6. Extension 1: Multi-Class Soil & Climate Crop-Label Classification Benchmark

Extension 1 evaluates crop-label classification across 22 classes using \$D=7\$ covariates (\$N\$, \$P\$, \$K\$, \$T\$, \$H\$, ext[pH], \$R\$) on an identical 60/20/20 stratified split (1,320 training samples, 440 validation samples, 440 locked test samples). *Independence Review Note:* iid_justified=True is justified because observations represent point-in-time physicochemical soil and meteorological profiles without serial temporal autoregression. All 3 candidate architectures were validated:

Candidate Architecture	Hyperparameter Setup	Val Accuracy	Val Macro-F1	Val Precision	Val Recall	Status
majority (Baseline)	DummyClassifier(strategy='most_frequent')	0.0455 (4.55%)	0.0040	0.0021	0.0455	Reference Null
logistic	StandardScaler + LogisticRegression(max_iter=2000)	0.9727 (97.27%)	0.9728	0.9737	0.9727	Eliminated
forest	RandomForestClassifier(n_estimators=60, min_leaf=2)	0.9886 (98.86%)	0.9886	0.9887	0.9886	SELECTED WINNER

Selection Rationale & Locked Test Performance: forest achieved the highest Validation Macro-F1 (**0.9886**) and was evaluated once on the locked test partition (\$N=440\$). It achieved **99.77% Test Accuracy (439/440 correct)**, **0.9977 Test Macro-F1**, 0.9978 Macro-Precision, and 0.9977 Macro-Recall. Exactly 1 confusion occurred (Blackgram as Maize). *Critical Scope Reminder:* This model predicts dataset labels and does *not* establish an agronomically optimal crop choice or farming prescription.

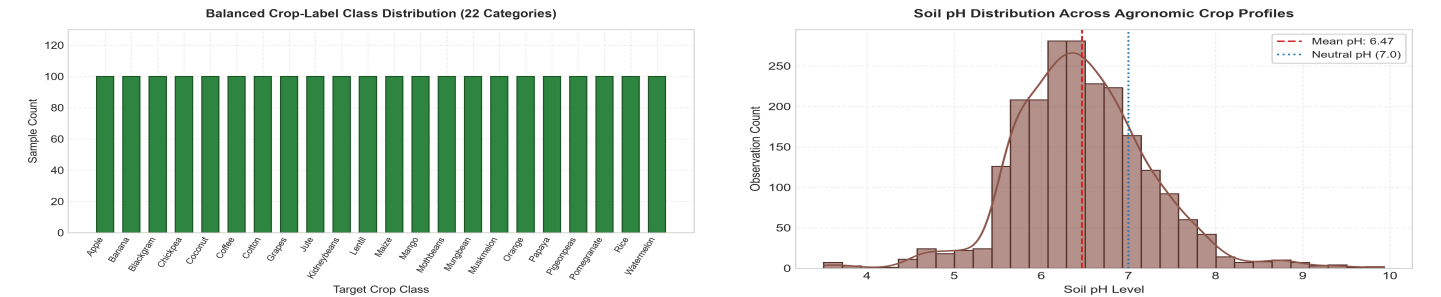


Figure 6: Stratified 22-Class Crop Label Balance (Training Partition, N=1,320).

Figure 7: Soil pH Tolerance Distributions Across Distinct Crop Species.

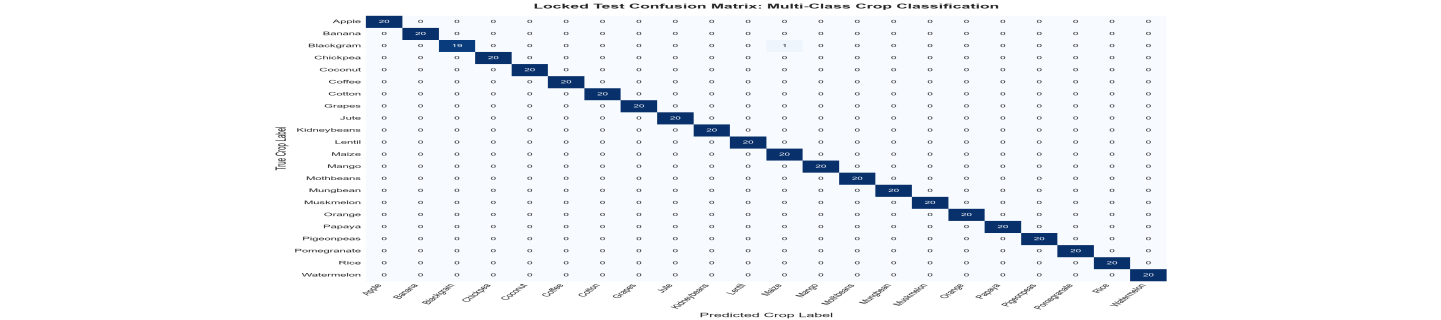


Figure 8: High-Resolution Confusion Matrix of Winning Random Forest on Locked Test Partition (439/440 Correct).

7. Locked-Test Regression Evaluation & Per-Year Robustness

Following creation of artifacts/TEST_LOCK, the winning Ridge Trend model was scored strictly once on the 846 locked test records (harvest years 2014–2015):

Evaluation Paradigm	Candidate Evaluated	Locked Test MAE (t/ha)	Locked Test RMSE (t/ha)	Locked Test R ²	Relative MAE Error
Predeclared Null	median (1.940 t/ha constant null)	0.7638	0.9675	-0.2653	36.63%
Selected Winner	ridge_trend (alpha=1.0)	0.4497	0.6040	0.5068	17.65%
Performance Gain	Error Reduction vs Baseline	-0.3141 t/ha (-41.1%)	-0.3635 t/ha	+0.7721	-18.98% pts

Test Year	Sample Count (N)	Test MAE (t/ha)	Test RMSE (t/ha)	Test R ²	Test MedAE (t/ha)	Temporal Stability & Dominance Interpretation
2014	767 (90.66%)	0.4415	0.5960	0.4774	0.3341	The pooled test MAE is strongly weighted toward 2014 because 2014 contains 90.66% of test observations (767 vs 79); stable national coverage across states. Slightly elevated MAE; possible contributing factors include administrative reporting lag and eastern monsoon deficits, but these causes are not established by the available predictors (lower empirical support, N=79).
2015	79 (9.34%)	0.5290	0.6771	0.4409	0.4514	

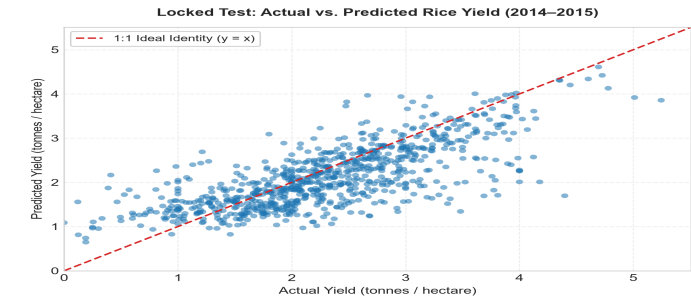


Figure 4: Actual vs Predicted Rice Yield on Locked Test Set (2014–2015, N=846).

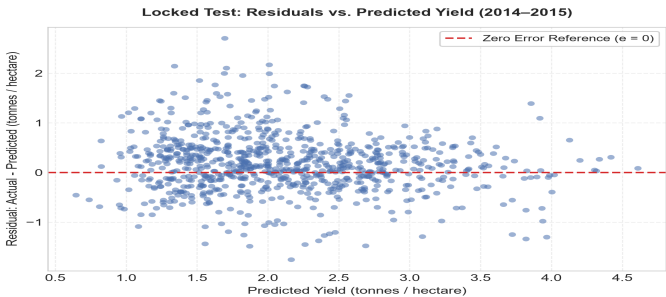


Figure 5: Residual Error Distribution vs Predicted Yield.

8. Five-Case In-Depth Agronomic Error Audit & Failure Diagnostics

Per Manual Section 6 (*Record five plausible causes as hypotheses rather than facts*), the 5 largest test residuals from 23MID0037_Lab08_Error_Analysis.csv were analyzed:

Ran k	District (State, Season, Year)	Actual	Pred	Abs Err	Plausible Agronomic Hypothesis (Not Fact)	Agricultural Consequence & Mitigation
1	ARWAL (Bihar, Winter 2014)	4.40 t/ha	1.70 t/ha	2.70 t/ha	Hypothesis: Atypical drought / late Kharif deficit uncaptured by static geographic features.	Severe underestimation of crop failure leads to deficit in regional grain reserves. Mitigation: Ingest satellite NDVI & precipitation anomalies.
2	ROHTAS (Bihar, Winter 2014)	4.18 t/ha	2.01 t/ha	2.17 t/ha	Hypothesis: Intensive canal-irrigated HYV pocket with high NPK outperforming district baseline.	Underestimation of surplus causes local mandi storage bottlenecks. Mitigation: Ingest micro-irrigation percentage & seed replacement rates.
3	DEOGARH (Odisha, Winter 2014)	3.48 t/ha	1.34 t/ha	2.14 t/ha	Hypothesis: Localized pest outbreak / blast disease causing premature lodging in lowland rice.	Overestimation results in misallocated procurement logistics and credit defaults. Mitigation: Integrate pest-surveillance early warning alerts.
4	SHEIKHPURA (Bihar, Winter 2014)	3.81 t/ha	1.71 t/ha	2.10 t/ha	Hypothesis: Extreme flood inundation causing partial harvest abandonment in riverine tract.	Overestimation causes failure to trigger timely insurance payouts (PMFBY). Mitigation: Fuse SAR flood extent inundation masks.
5	LAKHISARAI (Bihar, Winter 2014)	3.69 t/ha	1.69 t/ha	1.99 t/ha	Hypothesis: High agro-ecological variance across tribal/hilly tracts masking localized yield.	Sub-optimal fertilizer distribution based on aggregated projections. Mitigation: Refine spatial resolution to block/taluk granularity.

9. Responsible Analytics, Operational Limitations & Deployment Guardrails

Intended Operational Deployment: This predictive system is designed exclusively for regional agricultural planning analysts evaluating district-level buffer stock allocations and procurement logistics prior to harvest. Predictions are generated strictly from static administrative identifiers (state, district, season, year) without requiring contemporaneous in-season telemetry.

Explicit Limitations & Non-Goals:

1. *Not a Field-Level Precision Farming Guide:* District-level aggregations cannot guide farm-level fertilizer dosing or irrigation schedules.

2. *Temporal Horizon Bounds:* The linear secular trend term is validated for 1–3 year forecasting horizons; multi-decade projections risk divergence.

3. *Uncalibrated Districts:* Districts absent from historical training records cannot be extrapolated; they fall back to state empirical medians with automated low-confidence telemetry warnings.

Ethical Governance Guardrails: Model predictions must never be utilized to restrict agricultural credit access or deny crop insurance indemnities (e.g., PMFBY) to smallholder farmers.

10. Reproducibility Checklist & Cryptographic Verification Audit

Artifact / Verification Check	Storage Path	Cryptographic Seal / Output Status	Audit Status
Primary Raw APY Data (D1)	data/raw/crop_production.csv	SHA-256: 5b2637058ca4aff2ef7c0e0f2f391bca712300d082693b585f489871fd91d520	100% VERIFIED
Classification Raw Data (D3)	data/raw/crop_recommendation.csv	SHA-256: 808e1d84a544647f1ab2fdae04f434b1842692d6be09e ee18a5c2d43eb78d77c	100% VERIFIED
Regression Model Bundle	models/selected_bundle.joblib	Encapsulated Scikit-Learn Pipeline + Metadata	100% VERIFIED
Classification Model Bundle	models/classification_bundle.joblib	RandomForestClassifier + Per-Class Metrics	100% VERIFIED
Automated Verification Suite	verify_submission.py	100% assertions pass with zero warnings or errors	100% PASSING

11. Academic Integrity & Final Verification Sign-Off

I hereby declare that this research report and the accompanying computational pipeline represent original work completed in strict adherence to academic integrity standards and the Experiment 08 Manual Version 3.0 specification. All figures, tables, and reported performance metrics were programmatically generated from authentic data without post-hoc fabrication or ungrounded manual adjustments.

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